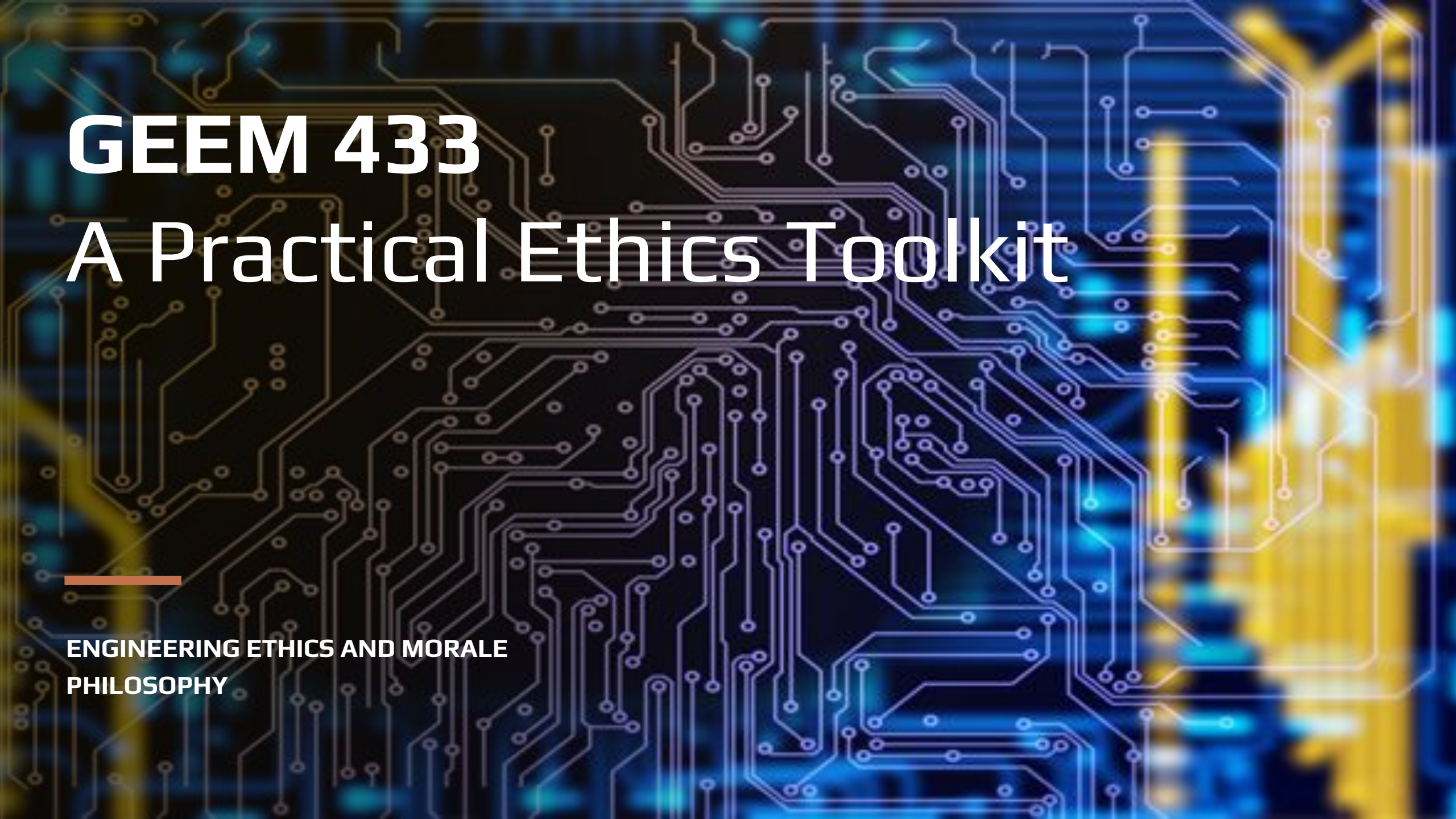




GEEM 433

A Practical Ethics Toolkit

**ENGINEERING ETHICS AND MORALE
PHILOSOPHY**



What is Philosophy?

Philosophy is an ancient discipline, which probably began as early as human civilization, that sought knowledge of all things on earth and in heaven. Knowledge of natural things and their causes lead to the creation of physics and metaphysics. Knowledge of human affairs and their constitutive relations lead to the creation of ethics, politics, and the philosophy of history. Knowledge of heavenly things leads to cosmology and speculative theology.

Philosophy in the ancient world was the parent of most scientific disciplines.

Philosophy

In order to understand philosophy you must not only grasp its subject matter, such as metaphysics and ethics, but also its method. In western philosophy the method to obtain knowledge is rooted in the philosopher's ability to form and evaluate arguments.

In Asian philosophy there is greater emphasis on knowledge of the Way (Dao) to live a life harmonizing the individual with her natural and social world. But in all cultures philosophy requires that we think critically: to be clear, precise, well-organized, truthful, complete, and able to handle objections. The study of critical thinking is called logic.

To be philosophical is to be a logical thinker who seeks knowledge of the whole. In this way philosophers avoid unsupported beliefs but base their views on good reason and evidence. Philosophers demand of themselves and others that they have reasoned logical belief.

Philosophy of Engineering

The philosophy of engineering is an emerging discipline that considers what engineering is, what engineers do, and how their work affects society, and thus includes aspects of ethics and aesthetics, as well as ontology, epistemology, etc

5 Concepts of Engineering Philosophy:

- Metaphysics
- Epistemology
- Ethics
- Logic
- Axiology

Metaphysics

At its core the study of metaphysics is the study of the nature of reality, of what exists in the world, what it is like, and how it is ordered. In metaphysics philosophers wrestle with such questions as:

- Is there a God?
- What is truth?
- What is a person? What makes a person the same through time?
- Is the world strictly composed of matter?
- Do people have minds? If so, how is the mind related to the body?
- Do people have free will?
- What is it for one event to cause another?

Epistemology

Epistemology is the study of knowledge. It is primarily concerned with what we can know about the world and how we can know it. Typical questions of concern in epistemology are:

- What is knowledge?
- Do we know anything at all?
- How do we know what we know?
- Can we be justified in claiming to know certain things?

Ethics

The study of ethics often concerns what we ought to do and what it would be best to do. In struggling with this issue, larger questions about what is good and right arise. So, the ethicist attempts to answer such questions as:

- What is good? What makes actions or people good?
- What is right? What makes actions right?
- Is morality objective or subjective?
- How should I treat others?

Logic

Another important aspect of the study of philosophy is the arguments or reasons given for people's answers to these questions. To this end philosophers employ logic to study the nature and structure of arguments. Logicians ask such questions as:

- What constitutes "good" or "bad" reasoning?
- How do we determine whether a given piece of reasoning is good or bad?

Axiology

Axiology is the branch of practical philosophy which studies the nature of value.

Axiologists study value in general rather than moral values in particular and frequently emphasize the plurality and heterogeneity of values while at the same time adopting different forms of realism about values.

- Axiology can be identified as the ultimate study of different materials with different values.
- The study tries to summarize different materials with value, nature, and characteristics.
- It also involves the study of principles and is divided into two major classifications, including ethics and aesthetics.

Axiology is one of the main issues in philosophy. There are two parts to axiology: ethics and aesthetics. The concept of axiology delves into the essence of value and the metaphysical aspects. The theory works on comprehending nature and arguing about what is actually valued. The conceptualization directly involves the objectives of the research.

Ethical Dilemmas

An ethical dilemma is a type of ethical issue that arises when the available choices and obligations in a specific situation do not allow for an ethical outcome. In such instances, a choice or an action is required and all of the available alternatives violate an explicit ethical principle or guideline.

An ethical dilemma is a paradox that comes up when there are two or more options, but neither of them is the best ethical or moral option. False accounting, sexual harassment, data privacy, nepotism, discrimination—are just some of the ethical dilemmas that happen in today's workplace.

- Engineers will likely have to deal with engineering ethical dilemmas as they develop new technology, engineering processes, and products. Though engineers may not always be directly responsible for these decisions, they still have a responsibility to do what's best for the public.
- They also have a responsibility to disclose their engineering decisions and explain why their choice was necessary or helpful. This is vital in engineering practice where there are often multiple choices that can all lead to negative consequences.

Ethical Dilemmas - Examples

#1 An ethical dilemma faced by the engineers designing an aircraft: lowering safety standards by designing an aircraft with less structural strength to make it easier to manufacture, or engineering an aircraft with engineering standards that will have high safety standards but make it more expensive and time-consuming to make.

#2 Ethical Dilemma faced by engineering design team developing a new car: engineering an aesthetically pleasing design with one-time convenience features that many consumers want but adding to the environmental costs of manufacturing, or engineering a product whose look is not as pleasing for most consumers but will have less impact on the environment.

Ethical Dilemmas

What are some of the ethical issues in computer science and technology?

- Use of artificial intelligence.
- Negative environmental impact.
- Misuse of personal and sensitive data.
- 'Deep fakes' and misinformation.
- Use of autonomous technology.